

Table 2. Baseline to 16-Week Changes in Objectively Measured Physical Activity and Body Weight

	Web-based tracking group (n=25)				Pedometer group (n=26)				Effect size (Cohen's d)
	Baseline	16 weeks	Change	p-value	Baseline	16 weeks	Change	p-value	
Minutes/week of physical activity									
Moderate to vigorous intensity (total)	172 (83)	234 (119)	62 (108)	0.008	176 (117)	189 (99)	13 (98)	0.51	0.48
Moderate to vigorous intensity (in bouts)	24 (39)	62 (82)	38 (83)	0.01	42 (68)	57 (69)	16 (76)	0.26	0.28
Light intensity	1,276 (311)	1,262 (320)	-14 (204)	0.49	1284 (383)	1,252 (317)	-33 (225)	0.82	0.09
Average steps per day	5,906 (1,968)	6,695 (2,708)	789 (1,979)	0.01	5,827 (2,431)	6,188 (2,423)	362 (1,605)	0.17	0.24
Body weight (kg)	82.4 (14.7)	82.2 (16.0)	-0.3 (2.4)	0.49	79.3 (12.2)	79.2 (13.2)	0.01 (2.3)	0.98	0.06

Discussion

In this study, a Fitbit-based intervention was associated with increased PA, with no change observed in the Pedometer Group. Although between-group tests did not reach significance, this may be due to sample size. The increase of 62 minutes/week of MVPA observed in the Fitbit arm is substantial, particularly if maintained over time. This maintenance would likely require progressive goal increases established via periodic contact with investigators via web or telephone. This study's findings differ from those of Thompson et al.,⁹ who found that a Fitbit with feedback did not increase PA among older adults. This may be partly because of the approximately 20-year difference in participant age.

Barriers to Fitbit use were low and technical issues were resolved quickly. Participants adhered well to wearing and viewing feedback on the tracker. Participant feedback indicated that women liked the website but either found the tracker sufficient for their needs or were otherwise unmotivated to explore the website beyond its basic functions; several mentioned the desire for additional hands-on training. Such training may help increase effect size by enhancing participants' engagement with the website.

Strengths of this study include the use of the Acti-Graph, detailed participant feedback, and use of Fitbit data to corroborate adherence. Limitations include a small sample, short duration, and lack of generalizability. Further research is needed to better understand the potential of direct-to-consumer devices for PA promotion, particularly over the long term.

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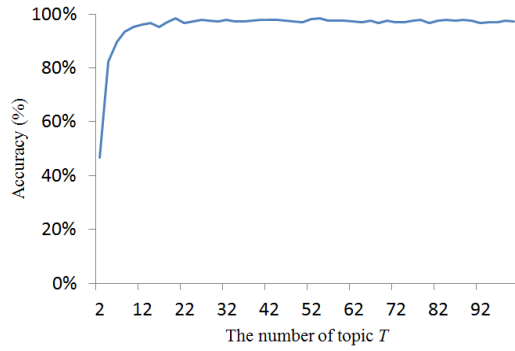


Figure 4. The influence of the number of topics T when $K=30$.

TABLE II. COMPARISON OF OPTIMAL RECOGNITION PERFORMANCE OF THE STATE-OF-THE-ART METHODS AND OUR PROPOSED METHOD

Method	Accuracy (%)
Our proposed method	98.41
Time-Frequency Feature [8]	88.57
HOD [10]	91.48
SMIJ [11]	93.54

IV. DISCUSSION

In this paper, we presented a real-time human motion recognition system, which consists of feature extraction, codebook generation, motion-word representation, topic modeling and motion recognition. Here we discussed three state-of-the-art methods, as shown in Table II. Using common time-frequency feature [8] could achieve good recognition performance, while it is easy to result in "dimension curse". Gowayyed et al. [10] used HOD to describe the motion capture data. The HOD is constructed by counting the length of each displacement in an orientation angles histogram. However, the method put the focus on all skeletal data, which increased computational complexity and imported noise. Ofli et al. [11] used sequence of the most informative joints (SMIJ) as features. The joints are selected by highly interpretable measures, including the mean of joint angles, the variance of joint angles and maximum angular velocity of joints. In comparison with HOD, this method efficiently decreased computational complexity and reduced noise by picking out some important joints, so the accuracy is greater than that of HOD. However, the SMIJ features ignored the effect of motion speed.

In comparison to three methods, our approach extracted 6 most informative angle features to describe human motion in advance. It contributes to eliminate redundant information in the original motion capture data. Then we applied LDA to get the latent topic, which has more semantic information. According to the topic distribution, the pattern for each motion can be obtained accurately. Meanwhile, the selection of appropriate number of topics could degrade the computational complexity. Hence, our method achieved the best recognition performance.

This study has limitations. On the one hand, the orientation of human is hard to be determined. On the other hand, this study mainly concentrates on the movement of lower limbs. Our future work will focus on how to determine

the orientation of human and extend the study into more complex motions.

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Analysis of the brain activated distributions in response to full-body spatial vibrotactile stimuli using a tactile P300-based BCI paradigm

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Abstract— We report a comparison study of somatosensory evoked potentials (SEP) and classification accuracies in three electrodes (Pz, P3 and P4) on the primary somatosensory cortex (S1), which are associated with the somatotopic arrangements, applying vibrotactile stimuli to user's arms and legs as a tactile P300-based BCI paradigm.

I. INTRODUCTION AND METHODS

The full-body tactile P300-based brain-computer interface (fbBCI) paradigm is a six-command BCI application which gives spacial vibrotactile stimulus patterns to user's whole body parts [1]. Large transducers are applied to arms (stimulus pattern #1 & #2), shoulder (#3), waist (#4), and legs (#5 & #6) in order to evoke and analyze somatosensory evoked potentials (SEP). The objective of the current study is to reveal a correlation between brain activity distributions on the primary somatosensory cortex (S1) and vibrotactile stimulus patterns for arms and legs. Penfield & Rasmussen reported the somatotopic arrangement that the senses of body functions are distributed on the surface of S1 [2]. Our hypothesis is that a tactile P300-based paradigm can utilize the theory for SEP classifications by attaching electrodes to Pz, P3 and P4 where S1 is located. In the fbBCI paradigm, Pz corresponds to stimulus pattern #5 and #6 (both legs), whereas P3 and P4 to pattern #1 and #2 (both arms) based on the somatotopic theory [2]. In the presented study stimulus pattern classification accuracies and SEP brainwaves in Pz, P3 and P4 are compared in terms of fbBCI stimulus patterns, in other words, stimulated body areas. Ten healthy BCI-users (mean: 21.9) participated in the fbBCI EEG experiment. A g.USBamp (g.tec, Austria) was employed to capture SEP brainwaves, in which the sampling frequency was set at 512 Hz. The experimental sessions were executed five times for each user. The captured SEP brainwaves were conducted ten times off-line classification using the simple moving average (SMA) and non-linear SVM with Gaussian kernel.

II. RESULTS, DISCUSSION AND CONCLUSION

The mean SEP brainwaves of participated ten users in Pz, P3 and P4 were depicted in Figure 1. We assumed that mean SEP of P3 and P4 where mid-parietal on S1 were stronger than SEP of Pz where peripheral on S1 when stimulus pattern #1 and #2 were given to users, according to the somatotopic theory, and we assumed the opposite (Pz > P3 and P4 in pattern #5 and #6) as well. However, the significant differences between those SEPs were not confirmed especially in nearby P300 peaks as shown in gray-shaded area of Figure 1. The somatotopic theory was not reflected on SEP brainwaves in fbBCI paradigm.

The classification accuracy results in Pz, P3 and P4 calculated by non-linear SVM were summarized in Table 1. The Pz scored higher mean accuracy (56.63%; averaged over blue-shaded cells) than the mean of P3 and P4 result (55.77%) in pattern #5 and #6, whereas mean of P3 and P4 scored higher mean accuracy (66.68%; orange-shaded cells) than mean Pz result (66.43%) in pattern #1 and #2. Those differences were slightly, though, it revealed the classification accuracy results were associated with the somatotopic arrangement, as the pattern for each body part scored higher accuracy result.

The presented results implied a possibility applying the somatotopic arrangements to SEP classifications in a BCI study. In the future study another electrodes on S1 (P1, P2 and P5, ... , P10) will be investigated for further development.

TABLE I. CLASSIFICATION ACCURACY RESULTS OF EACH STIMULUS PATTERN IN ELECTRODE CHANNEL PZ, P3 AND P4.

Stimulus pattern	Electrode channels		
	Pz	P3	P4
#1 (Left arm)	67.96 %	64.08 %	63.27 %
#2 (Right arm)	64.90 %	73.67 %	65.71 %
#5 (Left leg)	59.39 %	63.47 %	54.29 %
#6 (Right leg)	53.88 %	56.33 %	48.98 %

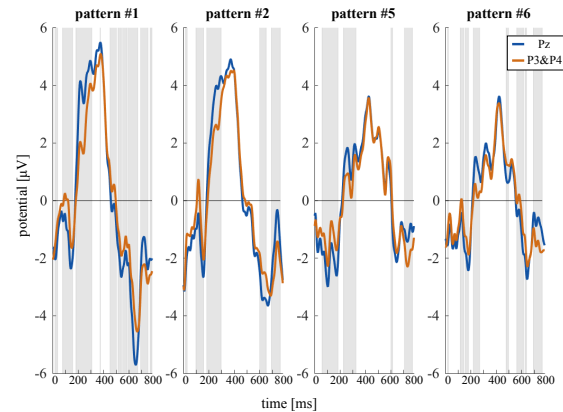


Figure 1. The mean SEP brainwaves of ten fbBCI users with electrodes Pz, P3 and P4. The blue line signified the SEP of Pz, whereas orange depicted mean SEP of P3 and P4. The gray-shaded area denoted the significant difference ($p < 0.01$; Student t -test) between two SEPs at each moment.

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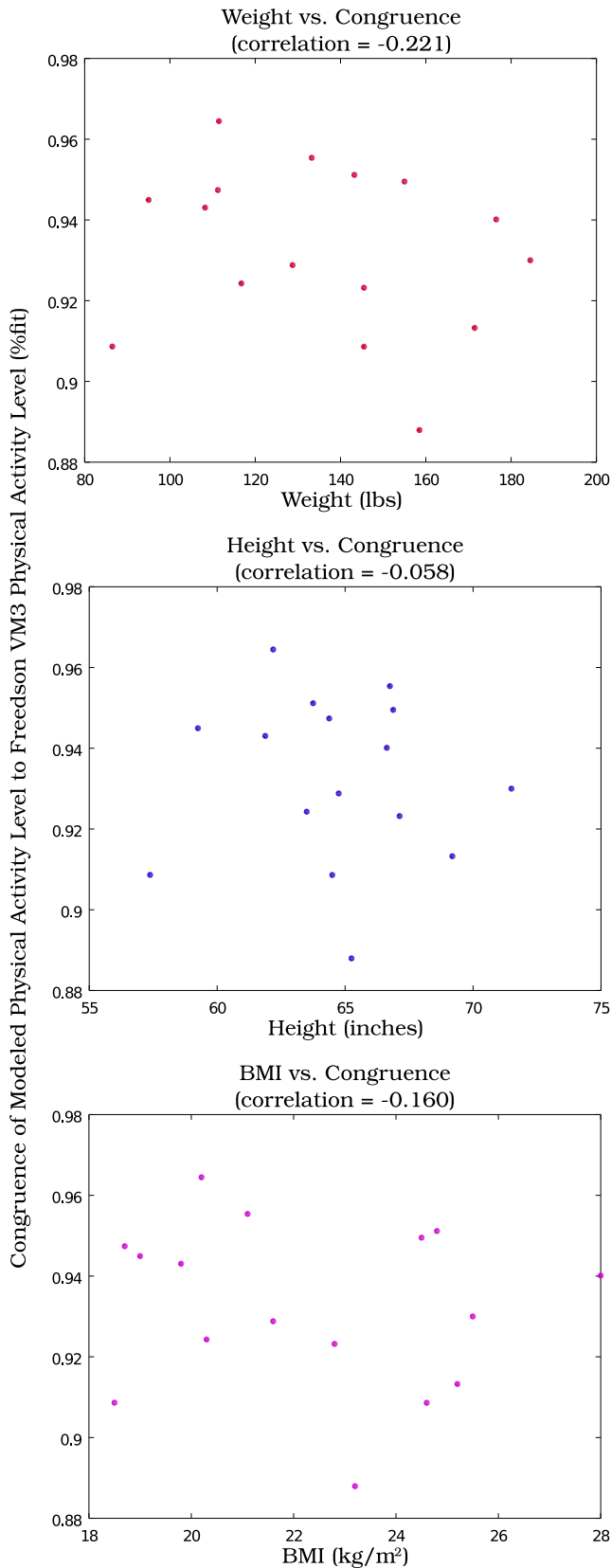


Fig. 4. This set of figures demonstrates how weight, height, and BMI have no observable effect in the overall accuracy of the model. Each of these factors presents a very low correlation to the model congruence, suggesting that the model congruence is independent of the subject's weight, height, or BMI.

Fitbit intensity and ActiGraph Freedson PA level was 80%, a rate unacceptable for clinical use of the Fitbit in place of the ActiGraph. However, by application of the modeling presented in this paper, that average congruence is increased to 93%, a point which is arguably within an acceptable range for use of the Fitbit in some studies.

VI. DISCUSSION

The ActiGraph and Fitbit each report activity level using very different processing algorithms. Through use of the Fitbit reported measures of steps, METs, and intensity level, one is able to predict with reasonable success the ActiGraph Freedson VM3 reported PA levels in 83.68% of the data collected in this study ($\Sigma(SSS, LLL, LML, MVM, <1\% \text{ agreements})$). Considering assessed PA levels of agreement between the Freedson VM3 and Fitbit only, the devices had a 80.03% agreement rate (this includes several categories not reflected in Fig. 3 due to the small size, $<1\% \text{ agreements}$). Most importantly, this model is able to correct the total time spent in moderate to vigorous PA levels (MVPA) to a measure no longer statistically significantly different from that which was reported by the ActiGraph Freedson VM3 model, a method considered to be the gold standard for PA assessment in the ecologically valid community setting. An over estimation of time spent in MVPA could be expected to be detrimental in determining the success of interventions. Given the data at hand, one is also able to estimate the errors in the model relative to the PA level assessed by the ActiGraph. This, shown in Fig. 5, also demonstrates that the model is most accurate for subjects who are less active, suggesting that the model would be well suited for use in applications where one is studying MVPA of largely sedentary subjects.

Not only have we outlined how to improve the MVPA estimates from the Fitbit Flex, we have also demonstrated the ability to collect data remotely. This contrasts with the traditional protocol where participants must return the data collection to the researcher for the researcher to manually download data from the GT3X device. Future longitudinal studies can leverage these tools that enable single in-person interactions with the participant, while continuously collecting measures of physical activity; participant contact can become minimized and can extend for the life of the device. The authors have developed the WearWare server system, utilizing the Fitbit API, and are in the pilot process of collecting data without third party engagements.

A. Limitations

One possible reason for the remaining differences between the ActiGraph and Fitbit is the placement of the device on the wearer; the Freedson VM3 algorithm was validated while ActiGraph device was worn at the waist. The Fitbit was worn on the wrist, and may be more susceptible to measurement error. However, the algorithms used in conjunction with ActiGraph have only been validated when the device is worn at the hip. Despite this measurement limitation, results of this study demonstrate the extent to which Fitbit-derived data can be improved, particularly MVPA, which is most often the primary outcome measure

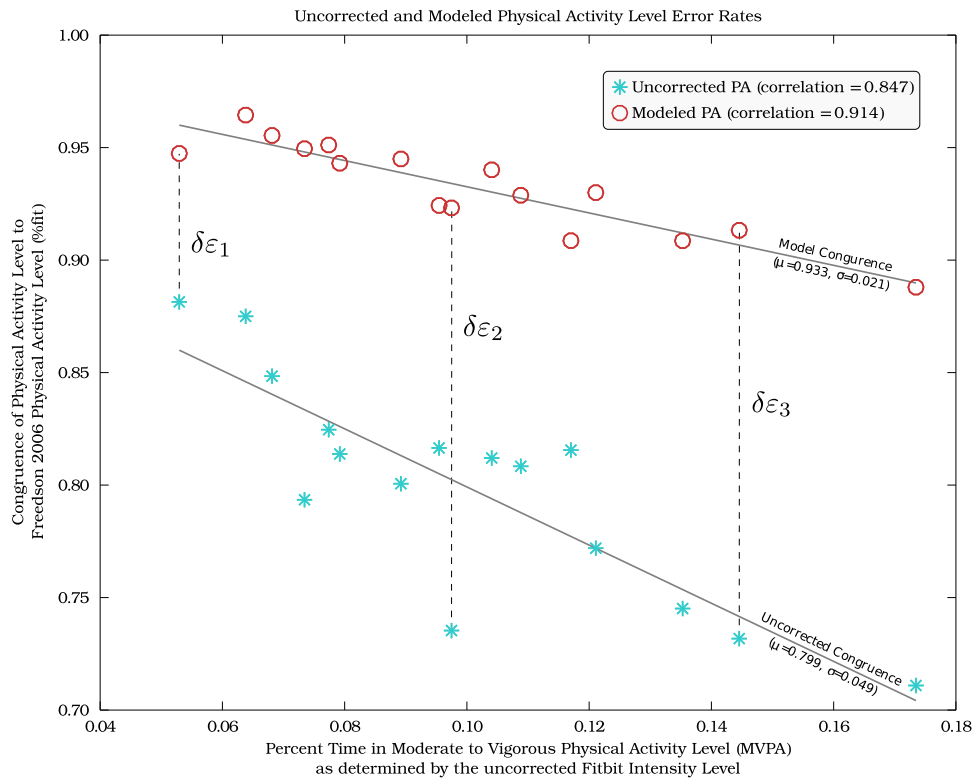


Fig. 5. Prior to modeling, the Fitbit was shown to have an error rate that substantially depended on the time spent in MVPA. At higher MVPA, these errors could be expected to be as great as 30%; 20% on average. However, with the application of the modeling, such error rates are reduced to 10% at most, or approximately 7% on average. One should also note the change in slope between the unmodeled and modeled lines here. While the modeled PA does still depend on the subject's time in MVPA, this dependence is not as strong as that established by the unmodeled PA. This is evidenced by the difference in slope, where a horizontal line would be reflective of no dependence of congruence on activity level. The degree of error reduction does vary between subjects, though as illustrated in Fig. 4, not predictably by height, weight, or BMI. Three examples are shown here, $\delta\epsilon_1$, $\delta\epsilon_2$, and $\delta\epsilon_3$. For each of these three subjects, the Fitbit estimated time in MVPA was dramatically different. The similarity of error reduction between $\delta\epsilon_2$ and $\delta\epsilon_3$, and difference between these and $\delta\epsilon_1$ demonstrates the variable reduction in error, specific to each participant.

used in interventions and epidemiological surveillance studies that assess PA.

Wearable activity monitors, including research-grade accelerometers are constantly evolving and most have now transitioned to being worn on the wrist given that wrist-worn devices are shown to result in greater wear-time adherence due to the convenience and unobtrusiveness of the devices.

The work presented here makes the assumption that the ActiGraph GT3X, in conjunction with the Freedson VM3 method, is able to accurately reflect the actual physical activity of the involved participants. Instead of relying on the measures assessed by the ActiGraph, future research should consider direct observation of the maximal oxygen consumption to establish participant physical activity level. A future study will also work to consider if other measures from the Fitbit, such as the recent advent of photoplethysmography (PPG) to monitor heart rate, can be leveraged to improve the estimates of PA using these low-cost commodity devices.

B. Conclusions

We now clarify the contribution of this work. Given the importance of measuring MVPA in physical activity research as an outcome variable and considering the fact that the majority

of literature supports the understanding that Fitbit overestimates PA, reliance on Fitbit reported PA estimates could be a detriment to determining the success of an intervention. The degree of error one should expect were outlined in Fig. 5. These errors approach 30% with physically active subjects. Errors are magnified as actual MVPA increases. These errors can potentially have major implications in interpreting MVPA outcomes if using a Fitbit without applying data corrections. For example, one could consider the case where MVPA increases a small amount as result of an intervention, but yet the reported MVPA increase is large, and thus an intervention might appear to have statistically significant impact when the reflected change is not actually large enough to trigger significance. The provided statistical model for correction of Fitbit PA assessment makes use of three measures provided by Fitbit, steps, METs, and Intensity Level, considering the relative probabilities of each Freedson based PA level assessment given the Fitbit measures, rather than creating a cut point like model from Fitbit data. Fitbit does not currently provide raw measures (the accelerations) from their devices, making such a cut point based algorithm unfeasible.

Finally, the authors would like to note that they have prepared the necessary functions to apply these techniques to existing Fitbit data as a library that can be used within Octave or Matlab. The details of this library, including how to request gratis access,